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$24i - 32$

$= (x + yi)^2 = x^2 - y^2 + 2xyi$

$$\left\{\begin{array}{lll} x^2 - y^2 & = & -32 \\ 2xy & = & 24 \end{array}\right. \Rightarrow \left\{\begin{array}{lll} x^2 - y^2 & = & -32 \\ xy & = & 12 \\ (xy > 0) & & \end{array}\right. \Rightarrow \left\{\begin{array}{lll} x^2 - y^2 & = & -32 \\ x^2(-y^2) & = & -144 \end{array}\right.$$

$\lambda^2 + 32\lambda - 144 = 0$

$D = 1600$

$\lambda_{1,2} = \dfrac{-32 \pm 40}{2}$

$\lambda_1 = 4$ en $\lambda_2 = -36$

$x^2 = 4 \Rightarrow x = \sqrt{4}$

$-y^2 = -36 \Rightarrow y = \sqrt{-36}$

Oplossing: $2 + 6i$ of $-2 - 6i$

References